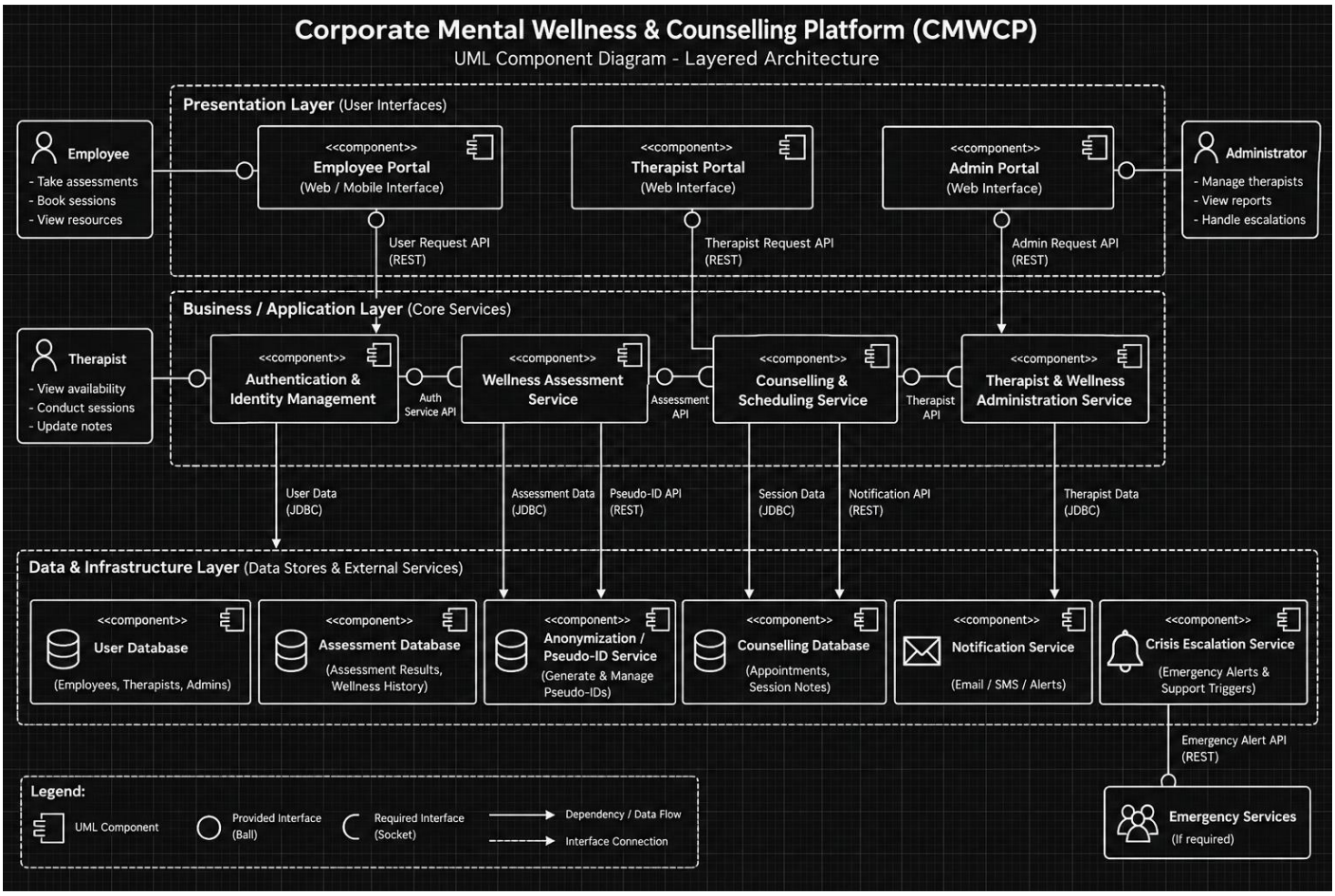
NAME: Vankadara Sri Kruthi **SRN:** PES2UG24CS574 **SEC:**5J

LAB 3 — COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Corporate Mental Wellness & Counselling Platform (CMWCP)

UML Component Diagram — Layered Architecture



Written Justification

Architecture Selection: We chose Layered Architecture for the Corporate Mental Wellness & Counselling Platform (CMWCP).

Architectural Choice

- Layered Architecture separates CMWCP into a Presentation Layer for Employee, Therapist, and Admin interfaces; a Business/Application Layer containing Authentication & Identity Management, Wellness Assessment, Counselling & Scheduling, and Therapist & Wellness Administration; and a Data & Infrastructure Layer containing databases and supporting external services.

Two Reasons

- Clear Separation of Responsibilities: Each layer has a defined role, making the platform easier to understand, develop, test, and maintain. Changes to user interfaces can be handled largely within the Presentation Layer, while business rules remain in the Application Layer.
- Maintainability and Controlled Data Access: Core services access the required databases and supporting services through defined interfaces. This keeps data handling organized and prevents the user interfaces from directly managing database operations.

Security Advantage

- Layered separation provides controlled access between the presentation, application, and data layers. Sensitive employee wellness, counselling, therapist, and assessment data is handled through application services rather than being directly exposed to the user interfaces. The Anonymization / Pseudo-ID Service further supports privacy by generating and managing pseudo-IDs.

Performance Benefit

- The separation of responsibilities allows application services and supporting data services to handle requests efficiently without placing all processing in the user-interface layer. Frequently used services such as Wellness Assessment, Counselling & Scheduling, and Therapist Administration can be optimized independently.

Conclusion

- Layered Architecture is appropriate for CMWCP because it provides clear separation of concerns, organized data access, better maintainability, controlled handling of sensitive information, and a structured foundation for the platform.